

Notes: Huberman (RFS, 2001)

Familiarity Breeds Investment

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1 Big picture

Question. Why do investors hold concentrated portfolios in familiar securities even when diversification is available? More specifically, do households buy the shares of companies they know because those firms operate in their local environment?

Setting. The paper studies the seven U.S. Regional Bell Operating Companies (RBOCs) in late 1996. Each RBOC is a large, NYSE-traded local telephone company with a geographically defined service area created after the 1984 AT&T divestiture.

Core challenge. The standard portfolio-theory benchmark says investors should diversify across securities and, if anything, avoid overloading on local firms because local firm payoffs may be correlated with local labor income, housing wealth, or regional business conditions. The observed choice to hold local RBOC stock therefore runs against a simple risk-sharing logic.

Main conceptual contribution. The paper argues that **familiarity is itself a determinant of portfolio choice**. Investors appear to treat familiar assets as more comfortable, safer, or more attractive, even when the familiar asset is not objectively better for diversification.

Main empirical design. Huberman compares, state by state, ownership of the local RBOC with ownership of the six out-of-state RBOCs. The key objects are the number of shareholder accounts, each RBOC's fraction of accounts in a state, each RBOC's fraction of equity held in a state, dollar amounts invested, and account sizes.

Main result on participation. In every RBOC-served state except Montana, more accounts hold the local RBOC than any other RBOC. The local RBOC also has a larger state-level fraction of its accounts than any out-of-state RBOC in every state.

Main result on dollars. In 44 of 48 states, the amount invested in the local RBOC exceeds the average amount invested in out-of-state RBOCs. In 26 of 48 states, the local RBOC receives more dollars than any other RBOC.

Main result on account size. Local holdings are not merely token positions. Excluding New York, the population-weighted average local RBOC account size is about \$14,400, while the comparable average for out-of-state RBOCs is about \$8,246. These holdings are large relative to typical U.S. household stock portfolios in the mid-1990s.

Economic takeaway. Familiarity helps explain several related puzzles: home-country bias, local bias in domestic portfolios, employee holdings of employer stock, and geographically concentrated investments. The RBOC evidence is powerful because all seven firms are traded in the same country and market, so usual international frictions do not explain the bias.

2 Data and measurement (key objects)

2.1 Regional Bell Operating Companies

The empirical objects are the seven RBOCs: Ameritech, Bell Atlantic, BellSouth, NYNEX, Pacific Telesis, SBC, and US West. Each company provides local telephone service in a particular region and is therefore familiar to households living in that region.

Interpretation. The RBOCs give a clean familiarity test. They are all large U.S. public firms and their stocks are accessible through the same national equity market, but customers are much more familiar with the RBOC that serves their own area.

2.2 Shareholder-location data

The RBOCs provide, by state, the number of accounts and the number of shares held in late 1996. Share prices are February 10, 1997 closing prices. The paper treats Washington, D.C. as a separate state-like observation and excludes Connecticut, Alaska, and Hawaii from the “local RBOC” comparisons because they were not served by an RBOC.

There are two important data qualifications:

- BellSouth reports only individual accounts registered directly with the company, excluding institutional and brokerage accounts.
- Ameritech allows its information to enter aggregate statistics but does not allow explicit publication of its state-by-state distribution.

Interpretation. The strongest results use account counts and account fractions, because those are less distorted by a small number of institutional accounts than are share-count or dollar-value measures.

2.3 Local RBOC indicator

Let s index states and j index RBOCs. Let $\ell(s)$ denote the RBOC that serves state s . A state has a local RBOC observation when $\ell(s)$ is defined.

$$\text{Local}_{sj} = \mathbf{1}\{j = \ell(s)\}. \quad (1)$$

Interpretation. The local indicator captures familiarity through the customer-service-area relationship: residents see, use, and know their local telephone company.

2.4 Number of accounts

Let A_{sj} be the number of shareholder accounts in state s holding RBOC j . The first participation measure is simply A_{sj} .

Interpretation. If familiarity matters, then $A_{s,\ell(s)}$ should be larger than the number of accounts holding nonlocal RBOCs.

2.5 Fraction of each RBOC’s accounts in each state

The paper also normalizes account counts by each firm’s total number of accounts:

$$F_{sj}^A = \frac{A_{sj}}{\sum_{s'} A_{s'j}}. \quad (2)$$

Interpretation. This controls for the fact that some RBOCs are more widely held overall. A high $F_{s,\ell(s)}^A$ means the local state is overrepresented among that RBOC’s owners.

2.6 Fraction of each RBOC’s equity held in each state

Let H_{sj} be the number of shares of RBOC j held in state s . The equity-share measure is

$$F_{sj}^H = \frac{H_{sj}}{\sum_{s'} H_{s'j}}. \quad (3)$$

Interpretation. This asks whether a disproportionate fraction of the local company’s shares are held in the local region. It is more sensitive to large institutional, brokerage, and trustee accounts, especially those with New York addresses.

2.7 Dollar amount invested

Let P_j be the share price of RBOC j . The dollar value of state s ’s holdings of RBOC j is

$$D_{sj} = P_j H_{sj}. \quad (4)$$

Interpretation. Dollar amounts combine familiarity with scale. They show whether the local company receives more money from local investors, not merely more accounts.

2.8 Average account size

Average account size is

$$AS_{sj} = \frac{D_{sj}}{A_{sj}}. \quad (5)$$

Interpretation. Account size measures whether the local RBOC position is economically meaningful. Large local account sizes imply that investors are not merely buying a symbolic share or two.

2.9 Descriptive facts

The paper emphasizes four descriptive facts:

- RBOCs are large and liquid public companies, with total market capitalization of about \$207 billion in the sample.
- The typical RBOC account is large: average account values in Table 1 range from about \$23,000 to \$45,000.
- RBOC service areas explain where shareholder accounts are located.
- New York is special because many institutions, custodians, brokerage firms, and trustee accounts use New York addresses; this makes account-count evidence cleaner than equity-share evidence.

3 Models and empirical specifications (all equations + interpretation)

3.1 Portfolio-theory benchmark

The benchmark portfolio-theory logic is simple: investors should hold diversified portfolios. If a household’s income and wealth are tied to the local economy, the local RBOC is not especially attractive as a hedge, because its payoff may be correlated with regional economic conditions.

Interpretation. Under this benchmark, one would not expect local households to overweight the local telephone company relative to otherwise similar out-of-state telephone companies.

3.2 Familiarity hypothesis

The familiarity hypothesis predicts that local investors overweight the local RBOC:

$$X_{s,\ell(s)} > \frac{1}{6} \sum_{j \neq \ell(s)} X_{sj}, \quad (6)$$

where X_{sj} can be account counts, account fractions, equity fractions, dollar holdings, or account size.

Interpretation. The prediction is directional and descriptive. The local RBOC should look more popular or more heavily held than the nonlocal RBOCs.

3.3 Comparison with the average out-of-state RBOC

For each state s , define the out-of-state average

$$\bar{X}_{s,-\ell(s)} = \frac{1}{6} \sum_{j \neq \ell(s)} X_{sj}. \quad (7)$$

The main comparison is

$$\Delta_s^{\text{avg}} = X_{s,\ell(s)} - \bar{X}_{s,-\ell(s)}. \quad (8)$$

Interpretation. $\Delta_s^{\text{avg}} > 0$ means that in state s , the local RBOC is held more intensely than the average nonlocal RBOC.

3.4 Comparison with the strongest out-of-state RBOC

The paper also compares the local RBOC with the most popular nonlocal RBOC:

$$X_{s,-\ell(s)}^{\max} = \max_{j \neq \ell(s)} X_{sj}. \quad (9)$$

$$\Delta_s^{\max} = X_{s,\ell(s)} - X_{s,-\ell(s)}^{\max}. \quad (10)$$

Interpretation. This is a tougher test. $\Delta_s^{\max} > 0$ means the local RBOC beats every out-of-state RBOC in that state.

3.5 State-level t-statistic

Tables 3 and 4 report a state-by-state t-statistic. Let m_s and SD_s be the mean and standard deviation of the relevant statistic across the six out-of-state RBOCs in state s , and let $x_s = X_{s,\ell(s)}$ be the local value. The paper reports

$$t_s = \sqrt{5} \frac{x_s - m_s}{SD_s}. \quad (11)$$

Under the null used by the paper, this statistic is compared with a t distribution with five degrees of freedom.

Interpretation. A large positive t_s means the local RBOC is not just slightly above the nonlocal average; it is far above the cross-RBOC variation among nonlocal firms in that state.

3.6 No causal regression or instrument

The paper does not estimate a causal regression and does not use an instrument. The design is a set of structured cross-sectional comparisons.

Interpretation. The empirical goal is not to estimate a treatment effect of familiarity, but to document a striking ownership pattern that is hard to reconcile with standard diversification logic.

4 Identification and instruments (what makes the evidence credible)

4.1 Why the RBOC setting is informative

The RBOC setting removes many standard explanations for home bias. All firms are U.S. companies, all are large, all trade on the NYSE, and all are accessible to U.S. investors. There are no international capital controls, currency barriers, or cross-border trading frictions.

Economic message. If local investors still overweight the local RBOC in this setting, the bias is unlikely to be mainly about legal or transaction barriers.

4.2 Why hedging does not explain the pattern

A customer might buy local RBOC stock to hedge future increases in local telephone prices. But household expenditure on local telephone service is too small to justify the observed stock positions. More important, regional risk-sharing logic points in the opposite direction: households whose economic fortunes are tied to a region should avoid adding more exposure to that region through the local RBOC.

Economic message. A pure hedging story has difficulty explaining the size and direction of local RBOC holdings.

4.3 Information versus familiarity

One alternative explanation is local information. Local customers may know more about the local telephone company. Huberman argues that information alone is incomplete. Superior information should sometimes generate selling or active trading, not merely static buy-and-hold local ownership.

Economic message. The evidence looks more like comfort, optimism, or affinity toward the familiar than like an aggressive information-based trading strategy.

4.4 Institutional and brokerage-account concern

Share-count and dollar measures are partly contaminated by institutional custodians and brokerage accounts. New York is the clearest example: it contains many institutional, custody, and brokerage addresses, so its share of equity held is much larger than its share of accounts.

Economic message. This concern is real, which is why account-count evidence is central. The account-count findings are extremely strong and less sensitive to large institutions.

4.5 What the paper can and cannot prove

The paper can show that local investors disproportionately hold local RBOC stocks. It cannot observe each household's full portfolio, cannot directly measure each investor's beliefs, and cannot fully separate familiarity from local information or loyalty.

Economic message. The evidence is best read as powerful descriptive evidence for familiarity-based investing, not as a fully identified causal estimate.

4.6 Relation to the literature

The paper connects several phenomena: international home bias, local bias among domestic money managers, employee ownership of employer stock, and concentrated holdings in locally visible firms. It also relates to Merton's incomplete-information model and to behavioral evidence from Heath and Tversky on perceived competence.

Economic message. Familiarity is a common mechanism linking several forms of underdiversification.

5 Tables and figures

5.1 Table 1: RBOC size and average account values

Read:

- The RBOCs are large public companies, with total market capitalization of about \$207 billion.
- Average account values are large, ranging from about \$23,000 for US West to about \$45,000 for Ameritech.
- The imputed S&P 500 portfolio sizes are in the millions of dollars, showing how concentrated an account would be if it represented a market-weighted position.

Economic message. These are not obscure local securities. They are large, nationally traded firms, which makes the geographic tilt in ownership more striking.

5.2 Table 2: State service areas and aggregate ownership geography

Read:

- The average RBOC line share across covered states is about 74.61%, and the phone-line-weighted average is about 76.96%.
- The table defines which RBOC is local in each state.
- New York stands out: it has 14.16% of RBOC accounts but 73.75% of RBOC equity, reflecting institutional and custody addresses.

Economic message. The table both defines familiarity geographically and highlights why account-count measures are cleaner than equity-share measures.

5.3 Table 3: Number and fraction of accounts

Read:

Table 1
End of 1996 properties of the RBOCs and holdings of their shares

Company	Ameritech	Bell Atlantic	Bell- South	NYNEX	Pacific Teleis	SBC	US West	Total
No. of shareholders (thousands)		834	867	1,088	866	671	797	722
Market capitalization (\$ billions)	38	31	46	23	18	35	16	207
Average account value (\$ thousands)	45	36	43	27	26	44	23	
Imputed portfolio size (\$ millions)	7.3	7.0	5.6	7.0	9.1	7.7	8.5	

Number of shareholders is from the companies directly. Market capitalization from companies SEC form 10Q for Q3 1996. The number of shareholders for BellSouth is the number of individuals who are registered directly with BellSouth (i.e., for BellSouth, institutional and brokerage accounts are excluded). Imputed portfolio size is the average account value divided by the corresponding fraction of S&P 500. (Fraction of S&P 500 is computed by dividing the RBOC's 2/10/97 market capitalization by \$6 trillion, the approximate value of the S&P 500 portfolio.)

Table 2
General statistics

State	RBOC	% lines by RBOC	% RBOCs accounts in state	% RBOCs equity from state
AK	N.A.	0.00	0.06	0.01
AL	BLS	80.75	0.80	0.15
AR	SBC	68.98	0.50	0.18
CA	PAC	79.70	8.45	2.57
CO	USW	95.70	1.49	0.42
CT	N.A.	0.00	1.61	0.42
DC	BEL	100.00	0.22	0.08
DE	BEL	100.00	0.34	0.11
FL	BLS	59.24	6.65	2.08
GA	BLS	83.70	1.84	0.35
HI	N.A.	0.00	0.13	0.03
IA	USW	66.36	0.82	0.25
ID	USW	74.93	0.19	0.05
IL	AIT	83.64	6.75	2.45
IN	AIT	62.88	1.28	0.39
KS	SBC	83.60	0.73	0.26
KY	BLS	57.93	0.64	0.15
LA	BLS	92.83	0.96	0.19
MA	NYN	99.91	5.52	1.34
MD	BEL	99.81	2.65	0.72
ME	NYN	83.80	0.68	0.19
MI	AIT	85.19	2.32	0.83
MN	USW	75.35	0.95	0.27
MO	SBC	75.94	2.13	0.82
MS	BLS	93.75	0.49	0.10
MT	USW	69.54	0.27	0.06
NC	BLS	49.77	1.79	0.45
ND	USW	70.80	0.09	0.02
NE	USW	80.75	0.55	0.15
NH	NYN	93.98	0.89	0.22
NJ	BEL	96.77	8.92	2.60
NM	PAC	79.70	0.50	0.14
NV	PAC	28.70	0.36	0.11
NY	NYN	89.82	14.16	73.75
OH	AIT	59.09	3.06	1.03
OK	SBC	83.46	0.82	0.26
OR	USW	66.79	0.75	0.21
PA	BEL	77.56	5.69	2.31
RI	NYN	100.00	0.55	0.14
SC	BLS	67.10	0.80	0.18
SD	USW	77.77	0.14	0.04
TN	BLS	80.47	0.97	0.21
TX	SBC	77.10	3.81	1.37
UT	USW	95.56	0.34	0.09
VA	BEL	75.82	2.30	0.67
VT	NYN	84.27	0.35	0.08
WA	USW	69.37	1.24	0.34
WI	AIT	66.84	1.44	0.52
WV	BEL	83.52	0.58	0.17
WY	USW	83.04	0.14	0.05
Average		74.61		
Phone line-weighted average		76.96		

- The local RBOC has more accounts than any other RBOC in 47 of 48 states served by an RBOC; Montana is the only exception.
- The number of accounts in the local RBOC is more than twice the average number in out-of-state RBOCs.
- The local RBOC's number of accounts is 63% larger than the next most popular RBOC.
- In every state, the fraction of the local RBOC's accounts located in that state exceeds the highest corresponding fraction for any out-of-state RBOC.

Economic message. This is the cleanest evidence in the paper: customers disproportionately become shareholders of the company that serves them.

Table 3
Number and fraction of accounts

State	Average # and % acct's for out- of-state RBOCs		Highest # and % acct's for out- of-state RBOCs		# and % acct's for local RBOC		<i>t</i> -statistic	
	#	%	#	%	#	%	#	%
AK	512	0.06	635	0.08	No local	No local	No local	No local
AL	4,964	0.63	5,298	0.66	RBOC	RBOC	RBOC	RBOC
AR	3,517	0.42	4,747	0.44	17,214	1.58	64,75	115.78
AZ	10,363	1.22	12,471	1.32	7,842	0.98	14,25	53.54
CA	65,912	7.70	77,753	8.62	14,700	2.04	8.13	21.20
CO	10,976	1.29	13,652	1.39	98,515	14.68	12,49	27.60
CT	13,417	1.61	17,005	1.87	21,500	2.98	16,22	57.17
DC	1,678	0.20	2,217	0.21	No local	No local	No local	No local
DE	2,550	0.31	3,650	0.34	RBOC	RBOC	RBOC	RBOC
FL	48,935	6.17	54,614	6.31	2,596	0.30	7.22	37.58
GA	11,559	1.46	12,349	1.50	4,762	0.55	8.81	35.29
HI	1,106	0.13	1,374	0.16	94,929	8.72	19,45	79.94
IA	6,243	0.73	7,677	0.79	38,168	3.51	63.18	136.91
ID	1,439	0.17	1,726	0.20	No local	No local	No local	No local
IL	50,962	6.13	65,375	6.53	RBOC	RBOC	RBOC	RBOC
IN	9,572	1.15	12,904	1.19	10,300	1.43	11.11	36.01
KS	5,124	0.61	6,564	0.68	2,700	0.37	19.13	28.42
KY	4,345	0.55	4,813	0.60	Ameritech	Ameritech	Ameritech	Ameritech
LA	6,516	0.82	6,964	0.87	Ameritech	Ameritech	Ameritech	Ameritech
MA	41,200	4.98	52,829	5.22	11,905	1.49	20.30	46.25
MD	19,595	2.35	27,858	2.56	11,410	1.05	38.67	40.31
ME	5,025	0.60	6,722	0.62	17,013	1.56	50.09	44.79
MI	17,206	2.06	23,263	2.15	75,297	8.70	12.06	61.37
MN	6,976	0.82	8,775	0.87	37,261	4.30	9.32	38.29
MO	14,902	1.78	18,887	1.88	9,385	1.08	10.27	68.67
MS	3,111	0.39	3,275	0.41	Ameritech	Ameritech	Ameritech	Ameritech
MT	2,101	0.23	5,039	0.46	13,400	1.86	14.34	48.75
NC	12,335	1.56	14,091	1.63	35,296	4.43	21.27	88.11
ND	669	0.08	829	0.08	9,720	0.89	66.27	78.45
NE	4,143	0.49	5,039	0.53	3,400	0.47	2.04	4.37
NH	6,646	0.80	8,722	0.83	1,400	0.19	18.55	67.40
NJ	69,861	8.42	90,986	8.94	7,200	1.00	13.60	38.69
NM	3,873	0.46	4,756	0.51	12,339	1.43	11.33	107.41
NV	2,915	0.34	3,375	0.40	102,135	11.78	5.89	26.41
NY	110,051	13.30	141,217	13.95	5,900	0.82	9.12	32.91
OH	23,358	2.79	31,851	2.93	3,496	0.52	5.62	11.33
OK	5,719	0.68	7,368	0.73	166,966	19.29	7.52	35.33
OR	5,741	0.68	6,853	0.82	Ameritech	Ameritech	Ameritech	Ameritech
PA	41,527	4.98	58,410	5.37	13,592	1.71	20.74	56.46
RI	4,040	0.49	5,399	0.50	9,400	1.30	15.07	17.16
SC	5,314	0.67	6,127	0.71	83,127	9.59	10.58	53.54
SD	1,000	0.12	1,181	0.13	7,742	0.89	11.33	143.17
TN	6,426	0.81	6,839	0.85	14,602	1.34	36.30	71.71
TX	26,376	3.15	34,307	3.38	1,900	0.26	19.70	43.38
UT	2,473	0.29	2,972	0.33	18,387	1.69	48.58	81.57
VA	17,225	2.06	25,645	2.36	64,431	8.09	20.93	65.25
VT	2,626	0.32	3,539	0.33	5,100	0.71	23.10	38.04
WA	9,323	1.10	11,261	1.28	31,277	3.61	7.40	21.87
WI	10,278	1.23	13,769	1.30	4,574	0.53	8.73	59.51
WV	4,337	0.52	6,233	0.57	16,500	2.29	16.80	25.16
WY	1,029	0.12	1,188	0.14	Ameritech	Ameritech	Ameritech	Ameritech
Average (including Ameritech)					7,715	0.89	7.89	27.79
					1,900	0.26	21.86	31.55
Average (including Ameritech)							19.08	50.89

The number of accounts is the basic statistic for the left column of each pair; the ratio between the number of accounts of an RBOC in a given state and the total number of accounts for that RBOC, in percent, is the basic statistic for the right column of each pair.

Table 4
Fraction of an RBOC equity and dollar amount held in a state

State	Average fraction and dollar amount		Highest fraction and dollar amount for out-of-state		Fraction and amount held for local RBOC		<i>t</i> -statistic
AL	0.14%	40	0.17%	60	1.78%	311	150.70
AZ	0.49%	121	1.08%	188	0.58%	95	0.63
CA	2.76%	673	6.21%	1,082	6.29%	1,109	4.72
CO	0.45%	112	1.04%	181	1.03%	170	4.12
CT	0.55%	129	1.38%	240	No local	No local	No local
DC	0.09%	20	0.22%	38	RBOC	RBOC	RBOC
DE	0.12%	27	0.32%	56	0.15%	46	2.01
FL	2.01%	547	2.39%	841	0.22%	67	2.22
GA	0.34%	92	0.39%	138	10.42%	1,817	63.80
HI	0.05%	11	0.11%	19	4.66%	812	197.76
IA	0.29%	71	0.65%	114	No local	No local	No local
ID	0.06%	14	0.12%	22	RBOC	RBOC	RBOC
IL	2.32%	512	5.77%	1,007	0.48%	80	2.22
IN	0.38%	84	1.01%	176	0.11%	19	3.43
KS	0.24%	53	0.55%	97	Ameritech	Ameritech	Ameritech
KY	0.15%	41	0.19%	66	Ameritech	Ameritech	Ameritech
LA	0.18%	51	0.29%	97	0.57%	196	4.83
MA	1.51%	356	3.72%	649	1.19%	208	74.16
MD	0.79%	174	2.07%	361	1.65%	288	61.65
ME	0.22%	52	0.53%	93	2.66%	635	2.38
MI	0.81%	183	1.97%	344	1.48%	459	2.50
MN	0.29%	73	0.63%	109	0.34%	81	1.73
MO	0.65%	146	1.57%	274	Ameritech	Ameritech	Ameritech
MS	0.10%	26	0.13%	39	Ameritech	Ameritech	Ameritech
MT	0.07%	16	0.15%	27	0.57%	93	3.34
NC	0.43%	118	0.50%	172	1.07%	187	109.92
ND	0.03%	6	0.06%	11	0.16%	27	4.53
NE	0.17%	41	0.37%	64	3.36%	586	96.09
NH	0.25%	60	0.63%	110	0.07%	11	4.68
NJ	3.06%	692	7.51%	1,310	0.32%	53	3.20
NM	0.17%	42	0.39%	68	0.44%	106	2.28
NV	0.13%	32	0.28%	48	0.12%	1,127	1.10
NY	63.46%	16,658	79.19%	24,768	0.24%	39	1.29
OH	1.35%	289	4.45%	776	0.18%	31	1.35
OK	0.22%	49	0.55%	96	18.62%	18,625	1.26
OR	0.24%	59	0.53%	93	0.64%	217	5.77
PA	2.20%	502	4.99%	869	0.40%	65	2.23
RI	0.16%	37	0.40%	69	4.90%	1,516	4.45
SC	0.18%	48	0.22%	70	0.28%	67	2.32
SD	0.06%	13	0.19%	33	1.49%	260	92.38
TN	0.21%	57	0.25%	86	0.08%	14	0.88
TX	1.15%	252	2.76%	481	5.83%	1,017	344.05
UT	0.09%	24	0.19%	44	1.13%	1,136	6.23
VA	0.76%	168	2.18%	380	0.19%	32	3.85
VT	0.10%	23	0.25%	43	0.15%	36	1.65
WA	0.39%	96	0.87%	152	0.74%	122	3.07
WI	0.44%	96	1.11%	194	Ameritech	Ameritech	Ameritech
WV	0.20%	44	0.55%	96	0.33%	104	1.78
WY	0.06%	15	0.16%	28	0.08%	14	0.82
Average (including Ameritech)							27.24
							9.68

The basic statistics are the ratio between the number of shares of an RBOC in a given state and the total number of shares for that RBOC, in percent (for the left column), and the number of shares of an RBOC in a given state and the 2/1097 share price.

5.4 Table 4: Equity fractions and dollar amounts

Read:

- In all 48 states, the local RBOC's local equity fraction is above the average out-of-state RBOC equity fraction.
- On average, the local equity fraction is 2.76 times the average out-of-state equity fraction.
- In 19 of 48 states, the local RBOC's local equity fraction exceeds the highest out-of-state equity

fraction.

- In 44 of 48 states, dollars invested in the local RBOC exceed the average dollars invested in out-of-state RBOCs; the average ratio is 3.15.
- In 26 of 48 states, the local RBOC receives more dollars than any out-of-state RBOC.

Economic message. The local bias is not just about the number of investors. It also shows up in money invested, though this measure must be interpreted carefully because of institutions and custodians.

5.5 Table 5: Average account sizes

Table 5
Average account sizes

State	Average account size for out-of-state RBOCs, w/o BLS	Average account size for BellSouth	Average account size for the local RBOC
AK	6,210	9,154	No local RBOC
AL	7,789	18,072	18,072
AR	8,247	14,268	17,600
AZ	10,669	15,086	6,472
CA	9,300	13,916	11,259
CO	9,235	13,286	7,894
CT	8,468	14,131	No local RBOC
DC	10,633	17,290	17,710
DE	8,694	15,348	14,082
FL	10,982	19,140	19,140
GA	7,845	21,276	21,276
HI	8,727	13,699	No local RBOC
IA	10,215	14,862	7,726
ID	8,832	12,614	6,863
IL	8,430	15,399	Ameritech
IN	7,194	13,607	Ameritech
KS	9,040	14,738	16,477
KY	9,127	18,192	18,192
LA	7,624	16,910	16,910
MA	7,497	12,288	8,431
MD	7,484	12,961	12,311
ME	9,060	13,797	8,624
MI	9,266	14,794	Ameritech
MN	9,753	12,461	6,976
MO	8,349	14,509	19,544
MS	8,323	19,193	19,193
MT	9,202	5,320	7,812
NC	9,347	19,154	19,154
ND	8,020	13,014	7,789
NE	9,078	12,745	7,363
NH	7,810	12,661	8,576
NJ	8,548	14,393	12,508
NM	9,833	14,386	6,689
NV	10,378	14,331	8,830
NY	187,053	13,158	111,548
OH	8,665	24,348	Ameritech
OK	7,263	13,059	15,973
OR	9,473	13,510	6,943
PA	11,092	14,881	18,242
RI	8,045	12,816	8,608
SC	8,852	17,838	17,838
SD	8,816	27,641	7,198
TN	8,607	55,306	55,306
TX	8,224	14,007	17,636
UT	9,134	10,984	6,178
VA	7,936	14,818	14,055
VT	7,532	12,185	7,914
WA	9,386	13,529	7,420
WI	7,877	14,107	Ameritech
WV	8,488	15,375	13,439
WY	14,854	12,658	7,310
Average (w/o NY, including Ameritech)	8,869	15,681	13,817
Population-weighted average (w/o NY)	8,246	15,173	14,400
Population-weighted average (with NY)	21,947	16,137	22,585
Median	8,727	14,268	13,747

The basic statistic is the product of the number of shares per account of an RBOC in a given state and the 2/10/97 share price.

Read:

- Excluding New York, the average local RBOC account size is \$13,817.
- The population-weighted average local account size excluding New York is \$14,400.
- The comparable population-weighted average for out-of-state RBOCs excluding BellSouth and

New York is \$8,246.

- The median local RBOC account size is \$13,747.

Economic message. Local RBOC investors are not merely buying tiny symbolic positions. These are meaningful single-stock holdings relative to typical household wealth and stock ownership.

5.6 Compact summary of the empirical evidence

Object	Finding	Interpretation
Number of accounts	Local RBOC beats all others in 47 of 48 states	Strong participation bias toward the familiar firm
Fraction of accounts	Local RBOC beats the highest out-of-state RBOC in every state	Result survives normalization by RBOC popularity
Fraction of equity	Local RBOC beats out-of-state average in all 48 states	Local ownership extends to share quantities
Dollar amount	Local RBOC exceeds out-of-state average in 44 of 48 states	Familiarity affects money invested, not just account openings
Account size	Local accounts average about \$14,400 population-weighted without NY	Holdings are economically meaningful

Table 1: Main empirical facts in Huberman (2001).

6 Short summary

- Huberman studies whether familiarity affects investment by examining where RBOC shareholders live.
- The setting is useful because RBOCs are all large U.S. exchange-traded firms, so standard explanations for international home bias are weak.
- The core empirical result is that investors disproportionately hold the RBOC that serves their own region.
- The strongest evidence is based on account counts and account fractions, which are less distorted by institutions than dollar holdings.
- Dollar holdings and account sizes show that local investments are not merely symbolic.
- The paper interprets these patterns as evidence that investors prefer familiar assets and often underdiversify as a result.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a strong geographic link between investors and the firms they know. RBOC customers are much more likely to hold the stock of their local RBOC than the stocks of other RBOCs, even though all RBOCs are accessible through the same national stock market.

7.2 Economic intuition

Investors do not treat stocks only as abstract risk-return bundles. They also attach comfort, competence, optimism, loyalty, and attention to familiar companies. These psychological or informational channels push them toward local and personally visible firms.

7.3 Takeaway

Familiarity breeds investment. The same force that draws people to their local telephone company helps explain home-country bias, local bias, employer-stock concentration, and other forms of underdiversification. In portfolio choice, familiar stories can become market-level forces.